

Math Pretest

Due: Friday, September 18, end of day

Instructions

Set aside two hours for this pretest.

Please don't use a calculator, a computer, or any references. This pretest is meant to give both of us a clear picture of which areas you should practice more, so it won't help you to look things up. It is graded for completion, not correctness: doing it thoughtfully is what earns the credit.

Give your answers as expressions. For example, if your final answer to a question is $\sqrt{10}$, you should leave it as $\sqrt{10}$. If you have something that can easily be simplified, like e^0 or $\ln(1)$, you should simplify it.

1. Write the equation of a line with slope -5 that passes through the point $(-2, 2)$.

Answer. Point-slope form: $y - 2 = -5(x + 2)$, i.e. $y = -5x - 8$. Check: $x = -2$ gives $y = 2$.
✓ □

2. Find the slope and concavity of $f(x) = x^3 - x$ at $x = 2$.

Answer. $f'(x) = 3x^2 - 1$ and $f''(x) = 6x$, so at $x = 2$ the slope is $f'(2) = 11$ and the concavity is $f''(2) = 12$ (concave up). □

3. Calculate the following derivatives:

a. $\frac{\partial f}{\partial y}$ for $f(x, y) = e^{3x \sin(ky)}$

Answer. $\frac{\partial f}{\partial y} = 3kx \cos(ky) e^{3x \sin(ky)}$. □

b. $\frac{df}{dt}$ for $f(x, y) = x^3 e^y$, with $x(t) = \cos(2t)$ and $y(t) = t^2 + 1$

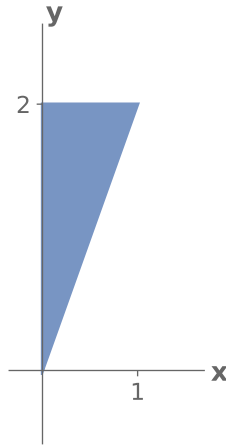
Answer. Chain rule: $\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = 3x^2 e^y (-2 \sin 2t) + x^3 e^y (2t) = 2t \cos^3(2t) e^{t^2+1} - 6 \sin(2t) \cos^2(2t) e^{t^2+1}$. □

4. Evaluate the following integrals:

a. $\int \sin(x) [2 \cos(x) + 3]^5 dx$

Answer. Let $u = 2 \cos x + 3$, $du = -2 \sin x dx$: $-\frac{1}{2} \int u^5 du = -\frac{1}{12} (2 \cos x + 3)^6 + C$. □

- b. Integrate $f(x, y) = x - y$ over the triangular region shown below.

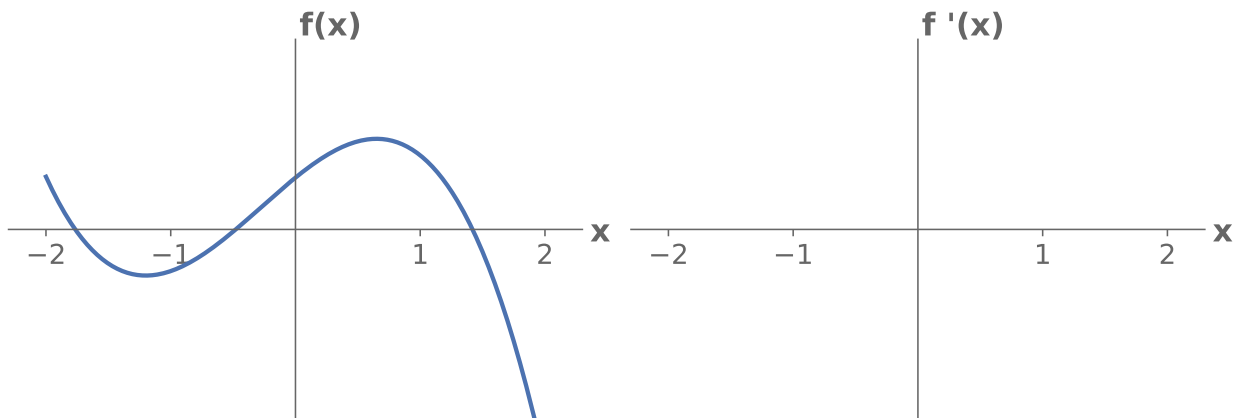


Answer. The slanted edge is the line $y = 2x$. Integrating y first: $\int_0^1 \int_{2x}^2 (x - y) dy dx = \int_0^1 \left[xy - \frac{y^2}{2} \right]_{2x}^2 dx = \int_0^1 (2x - 2) dx = -1$. (Integrating x first, $\int_0^2 \int_0^{y/2} (x - y) dx dy$, also gives -1 .) $\square$

5. The point P in the x - y plane has Cartesian coordinates $x = -2$, $y = -2$. What are its polar coordinates ρ and ϕ ? (You may have learned these as r and θ — they mean the same thing.)

Answer. Draw it: P is in the third quadrant. $\rho = \sqrt{(-2)^2 + (-2)^2} = 2\sqrt{2}$ and $\phi = \pi + \pi/4 = 5\pi/4$ (i.e. 225°). $\square$

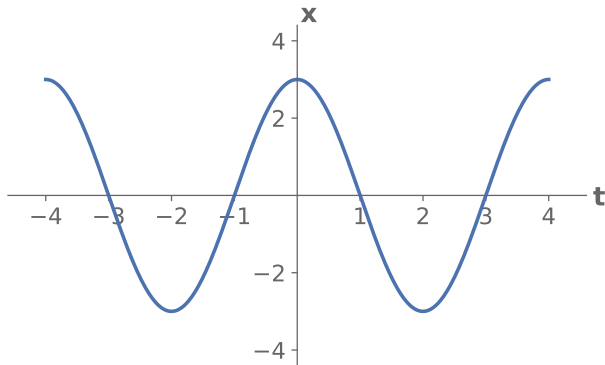
6. The plot below shows a function $f(x)$. On the axes provided at the right, sketch the function df/dx . Numbers are included on the horizontal axis, but you do not need to show numbers on the vertical axis.



Answer. Where f has its minimum (just before $x = -1$) and its maximum (between 0 and 1), $f' = 0$. To the left of the minimum and to the right of the maximum, f is decreasing, so $f' < 0$; between them f is increasing, so $f' > 0$. The sketch is a downward-opening arch crossing zero just before $x = -1$ and again between 0 and 1, negative and falling steeply at the right edge. $\square$

7. The plot below shows a cosine function $x(t)$.

- What are the amplitude A , period T , and angular frequency ω ?
- Write the function $x(t)$. *Hint: try plugging some points into your answer to make sure it matches the graph.*



Answer. $A = 3$ (the curve runs from -3 to 3); $T = 4$ (peak to peak); $\omega = 2\pi/T = \pi/2$. So $x(t) = 3 \cos(\pi t/2)$. Check: $x(0) = 3$, $x(1) = 0$, $x(2) = -3$, $x(4) = 3$. ✓ □

8. The vector $\vec{\mathbf{A}}$ is given by $\vec{\mathbf{A}} = -2\hat{\mathbf{x}} + 5\hat{\mathbf{y}}$. The vector $\vec{\mathbf{B}}$ has magnitude $2\sqrt{2}$ and direction 30° clockwise from the positive x axis.

- Find the magnitude and direction of $\vec{\mathbf{A}}$.

Answer. $|\vec{\mathbf{A}}| = \sqrt{4+25} = \sqrt{29}$. Draw it: $\vec{\mathbf{A}}$ points up and to the left, an angle $\tan^{-1}(2/5)$ counterclockwise from the positive y axis (equivalently $\tan^{-1}(5/2)$ up from the negative x axis). □

- Find the components of $\vec{\mathbf{B}}$.

Answer. $B_x = 2\sqrt{2} \cos(30^\circ) = \sqrt{6}$, $B_y = -2\sqrt{2} \sin(30^\circ) = -\sqrt{2}$ (negative: 30° clockwise is below the axis). □

- Find $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$.

Answer. $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = A_x B_x + A_y B_y = -2\sqrt{6} - 5\sqrt{2}$. Sanity check: the vectors point mostly away from each other, so a negative answer makes sense. □

9. An object is moving in some complicated way, and you want to solve for its acceleration at some particular moment. You can't actually do that calculation here, because we're not giving you enough information. Instead, check each of the following possible answers to determine whether it has correct units. Assume that:

- F_1, F_2 , and so on are forces,
- m is a mass,
- D is a distance,

- T is a time,
- v_0 and v_f are speeds.

Which of the following possible answers have correct units? (There may be more than one option with correct units.)

a. $a = \frac{v_0 + v_f}{T}$

b. $a = \frac{v_0 + v_f + 1}{T}$

c. $a = \frac{D}{T^2}$

d. $a = \frac{D}{T^2} e^{-v_f/v_0}$

e. $a = \frac{D}{T^2} e^{-v_f/(DT)}$

Answer. Acceleration needs units of distance over time squared. (a) speed/time ✓ YES. (b) NO — you can't add a speed and the unitless number 1. (c) ✓ YES. (d) the exponent is speed/speed, which is unitless, so ✓ YES. (e) NO — the exponent has units of $1/(\text{time})^2$, and exponents must be unitless. Correct units: a, c, and d. □